Year 10 Science (General)
Semester 1 Examination 2020 - Revision

Astronomy

1. Complete the following table.

TERM	DEFINITION
light year	Distance light travels in one year.
constellation	Group of stars forming a pattern.
galaxy	Collection of stars held together by gravity.
nebula	Gas cloud.
supernova	Massive explosion of a massive star near the end of its life.
red giant	Medium stars swell in size as they run out of fuel.
black hole	Very dense and huge gravity. Light can't escape.
sunspot	Dark areas on the surface of the Sun - sites of large eruptions.
corona	Upper atmosphere of the Sun.
fusion	Process producing energy in stars - hydrogen becomes helium.

2. Light travels at 300,000 km/sec. A star is 97.6 light years away. We can only travel at about 15 - 20 km/second with our best rockets.

(a) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer.

YES / NO (Circle your answer.)

Reasons:

- Spaceship is far too slow.
- The distance is huge and will take hundreds of years to reach.
- Will die well before we get there.

(b) Suggest ***one way*** we might be able to overcome the time factor of travelling to other galaxies, given it could be hundreds of years.

- Develop craft that can travel near the speed of light.
- Freeze people and wake them up hundreds of years later.
- Learn how to warp space and fold it on itself - makes the distance much shorter.

3. (a) What is the major element found in the Sun? hydrogen

(b) What happens to this element so that the Sun produces its energy?

Hydrogen atoms collide and become helium - needs 1 million °C to do this. Energy is released.

(c) The Sun loses 5 million tonnes of mass per second. According to Einstein's equation ($E = mc^2$), what does this mass become?

The lost mass becomes the energy that the Sun emits.

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
very hot	blue
cool	red
moderate	yellow

5. (a) About how old is our Sun (in millions of years)? 5000 million years

(b) How much longer will it "live"? 5000 million years

6. (a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → red giant → nova → white dwarf → black dwarf

(b) Do similar flow diagrams for the following types of stars.

Huge star → red supergiant → supernova → neutron star

Giant star → red supergiant → supernova → black hole

(c) The difference between a ***nova*** and a ***supernova*** is the size of the explosion that takes place. What happens to a star to cause it to explode?

Star starts to run out of fuel and (ultimately) collapses on itself.

7. (a) Describe the size of the gravity near a black hole.

huge

(b) Why does a black hole appear black?

Gravity is so big that light can't escape from it.

8. Complete the following matrix to compare the Big Bang and Steady State Theories, using ticks to show which statements apply to one, both or neither theory.

Statement	Big bang theory	Steady state theory
The universe has no beginning.		Y
The universe began with a single point called singularity.	Y	
The universe is expanding.	Y	
The universe always looks the same.		Y
The red shift in the spectrum of visible light coming from stars and galaxies provides evidence for the theory.	Y	
New stars and galaxies are created to replace those that move away due to expansion of the universe.		Y
This theory explains the amount of helium in the universe.	Y	
This theory is supported by the measurement of the current temperature of the universe (about –270 °C).	Y	
This theory was first supported by a Catholic priest.	Y	
The theory will never be proven incorrect.		
An end was put to this theory in 1965.		Y

9. Why are the constellations we see now so different from the way they were many centuries ago?

Stars have exploded at various times, so some have disappeared.

10. How have scientists gained their knowledge of the life and death of stars if the processes involved take millions of years to occur?

Light from distant stars takes millions of years to get to us, so we see them as they were at that time. Closer stars appear different, so we can see them at different stages.

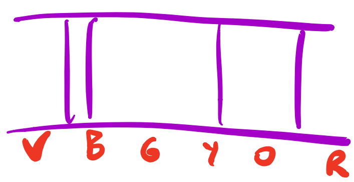
11. What is the difference between a ***neutron star*** and a ***black hole***?

Neutron star - remains of a massive star; dense; composed of neutrons.

Black hole - remains of a giant star; very dense; huge gravity; light can't escape.

12. What makes stars group together to form galaxies?

Gravity pulls all of the matter and stars together.



13. The Doppler effect is most commonly associated with the changing pitch of a sound as its source moves past you. For example, the pitch of the noise made by a speeding train increases as it approaches you and decreases as it moves away from you. Explain how the Doppler effect is relevant to the study of the universe.

The spectra of light from stars contain dark lines caused by the absorption of certain colours by substances in the star.

As the stars move away our solar system at high speed, the Doppler effect causes the lines to move towards the red end of the spectrum.

As the stars move towards our solar system at high speed, the Doppler effect causes the lines to move towards the blue end of the spectrum.

14. What is ***cosmic microwave background radiation*** and why does it exist?

Cosmic microwave background radiation is radiation that was released 380 000 years after the Big Bang.

15. Below is a list of the four spheres on Earth. Give a brief description of each.

Biosphere: Made up of all living things - plant and animals.

Atmosphere: Made up of the troposphere (lower atmosphere) and stratosphere (upper atmosphere). Contains the ozone layer that protects the planet from UV rays.

Hydrosphere: Made up of all the water on Earth, including that frozen in ice, liquid as water in oceans and contained in the atmosphere.

Lithosphere: Made up of the rocky crust and soil on the Earth.

16. (a) What is the ozone layer and why is it important for Earth?

A layer in the stratosphere that protects the surface from extreme UV radiation from the Sun.

(b) How has humankind damaged this layer?

Chlorofluorocarbons (CFCs) used as coolants and propellants in cans have damaged it, increasing the amount of UV reaching the surface.

17. What is the difference between ***weather*** and ***climate***?

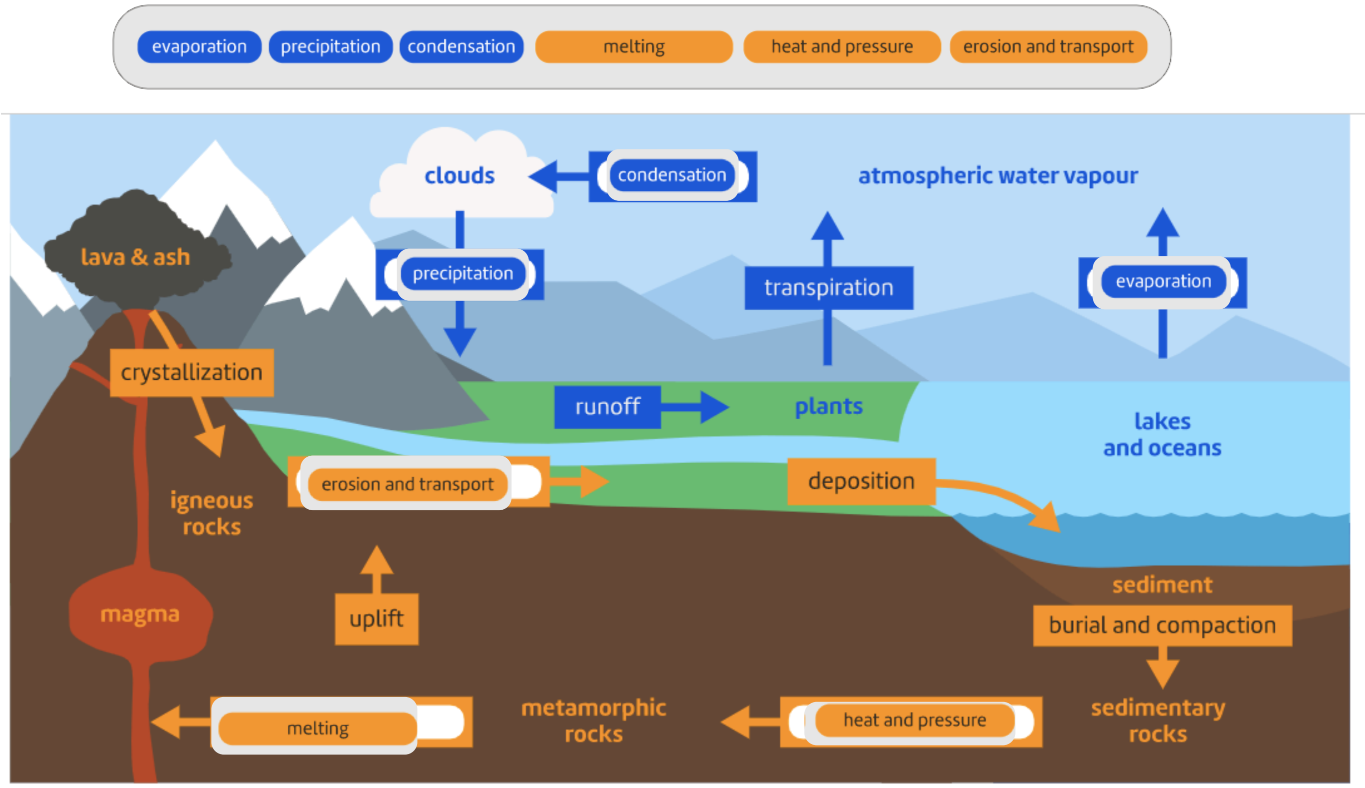
Weather: short-term changes in the atmospheric conditions, generally changing from day to day.

Climate: long-term observations of the atmospheric conditions, generally over many years.

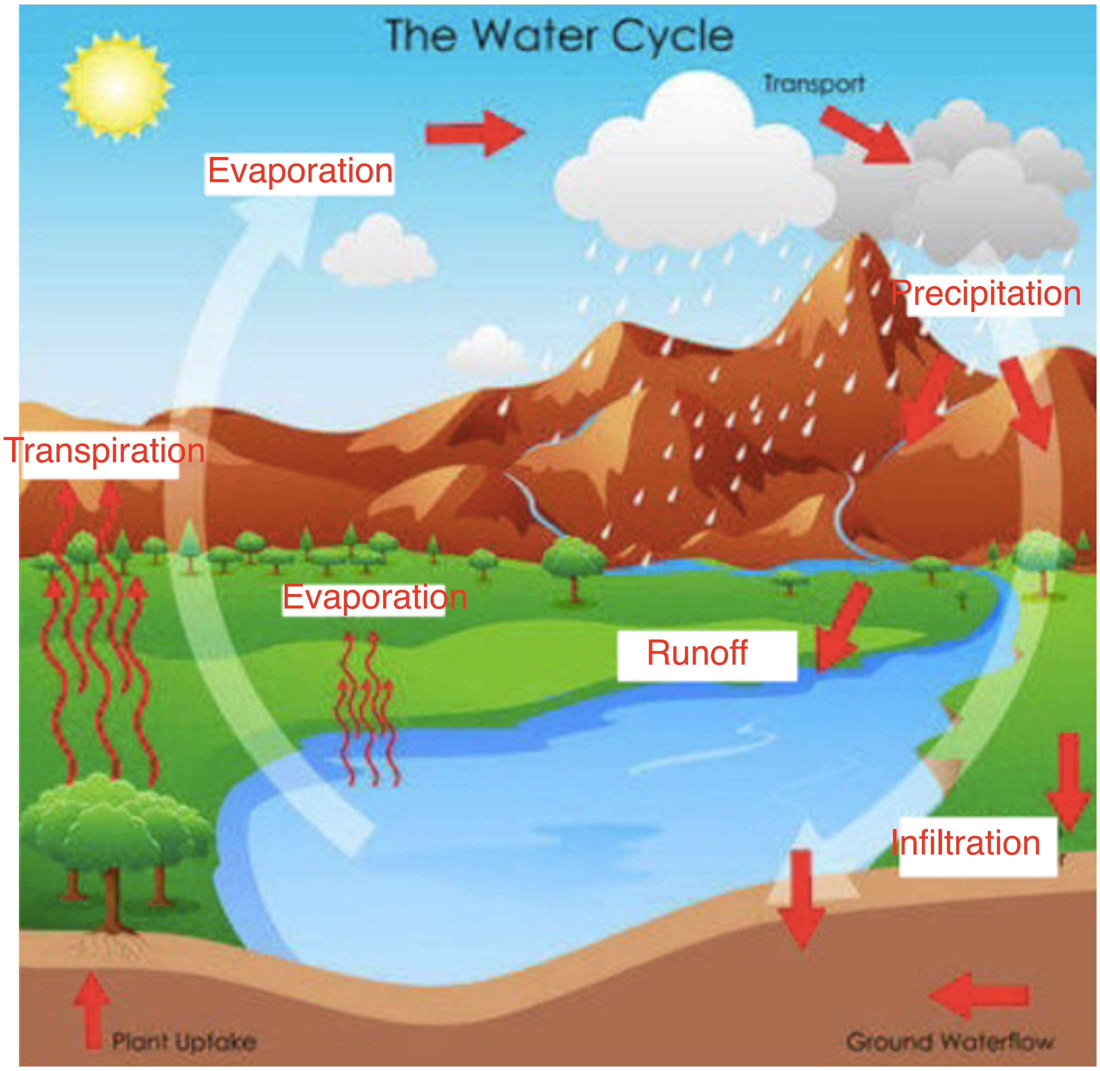
18. Describe two consequences of ***global warming***.

- Rising sea levels. Thawing of permafrost (frozen ground). Species extinction due to habitat change.
- Melting ice caps. Melting glaciers. More droughts, fires, etc.

19. The following diagram shows the rock cycle and the water cycle. Complete the diagram with appropriate labels.



20. (a) Label the diagram below showing the water cycle.

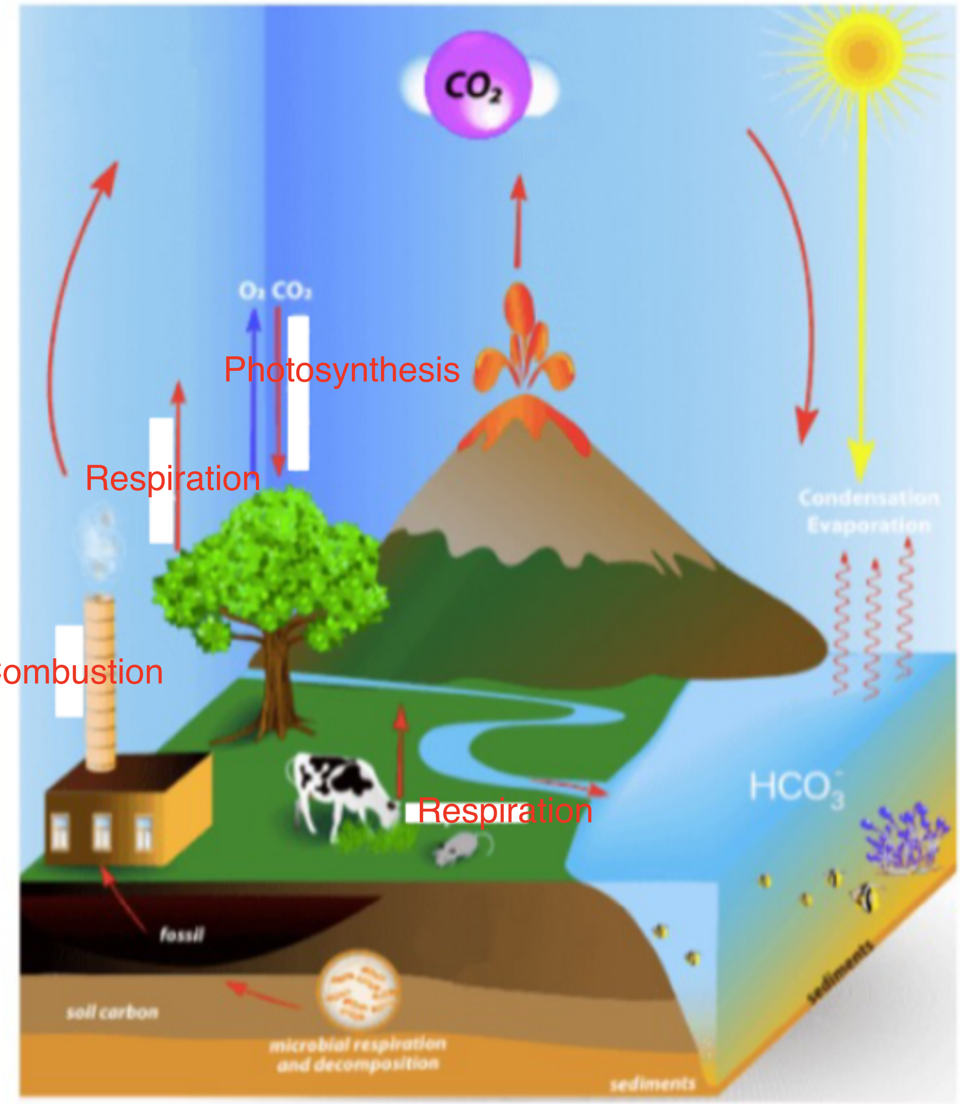


(b) Describe how water entering the roots of the tree ends up in the lake or river shown.

- Water taken up by roots and moves out into the atmosphere by transpiration.
- Water condenses into a cloud and eventually rains (precipitates) onto the ground.
- Water runs across the ground or flows underground into the lake.

21. The following diagram shows the carbon cycle.

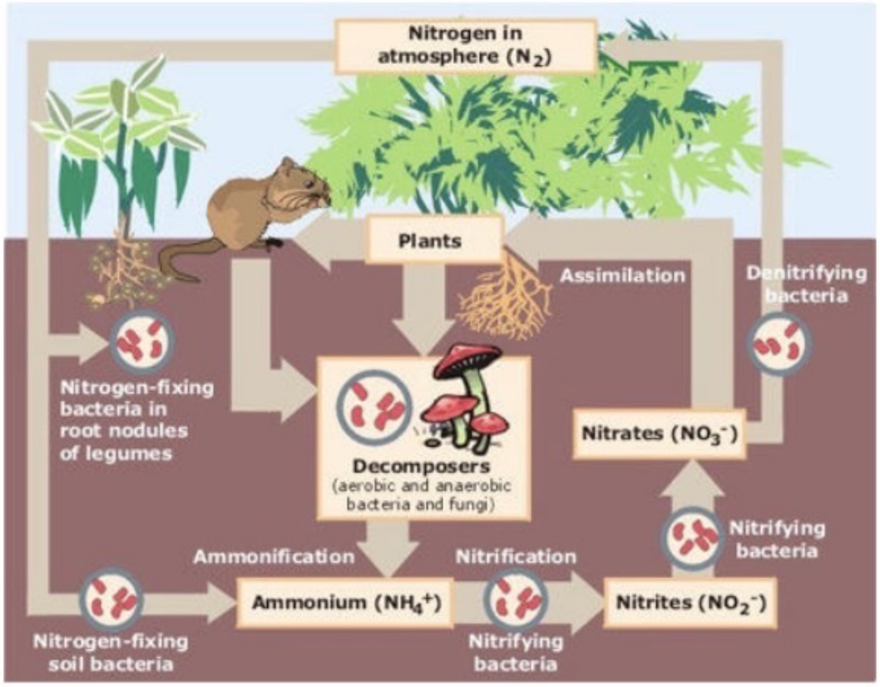
(a) Label the processes missing from the diagram.



(b) Describe how the carbon in the fossil is recycled to end up in the tissues and systems of the cow.

- Fossil fuel is burned in the power station and carbon dioxide is released into the atmosphere.
- Plants (grass) photosynthesise using the carbon dioxide and produce food for themselves.
- The cow eats the grass and the carbon is incorporated into its systems.

22. Use the diagram of the nitrogen cycle to answer the following questions.



(a) Explain how nitrogen gas (N_2) in the atmosphere is converted into ammonium (NH_4^{1+}).

N₂ is converted by nitrogen-fixing bacteria in nodules on roots and nitrogen fixing soil bacteria.

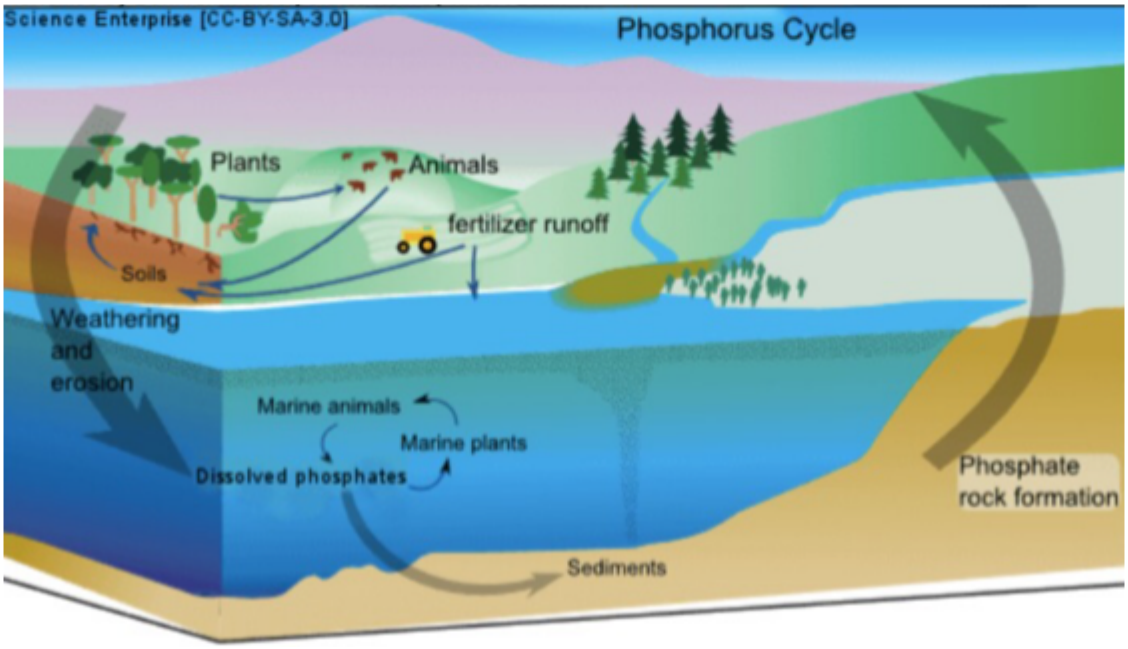
(b) Plants needs nitrates (NO_3^{1-}) to grow. How does NH_4^{1+} become NO_3^{1-}?

Nitrifying bacteria convert it to nitrites (NO2-) and then into nitrates (NO3-).

(c) What is the role of the decomposers in the cycle?

Break dead material down into ammonium (NH4-) in the soil.

23. The phosphorus cycle is shown in the following diagram.



(a) How does the phosphate in the rock become phosphate in the soil for plants to use?

By weathering and erosion, the phosphate moves into the soil.

(b) How does phosphorus get from the soil into an animal's systems?

- Phosphate is taken up by the grass as a fertiliser and grow.
- Animals eat the grass and the phosphorus gets incorporated into their systems.

(c) What is the role of marine plants and animals in recycling phosphorus?

- They use the dissolved phosphates in the marine environment.
- When they die, they become part of the sediments that form the rocks containing phosphates.

Human Biology

1. (a) What are the male and female ***gametes*** in animals called?

sperm and egg (ovum)

(b) What are the ***genotypes*** for a male and a female?

male XY female XX

2. DNA is thought to be a twisted ladder structure called a double helix. There are a series of bases located between the two strands of the helix.

(a) Name the four bases.

guanine (G) thymine (T) adenine (A) guanine (G)

(b) Which bases are paired together?

A - T
G -C

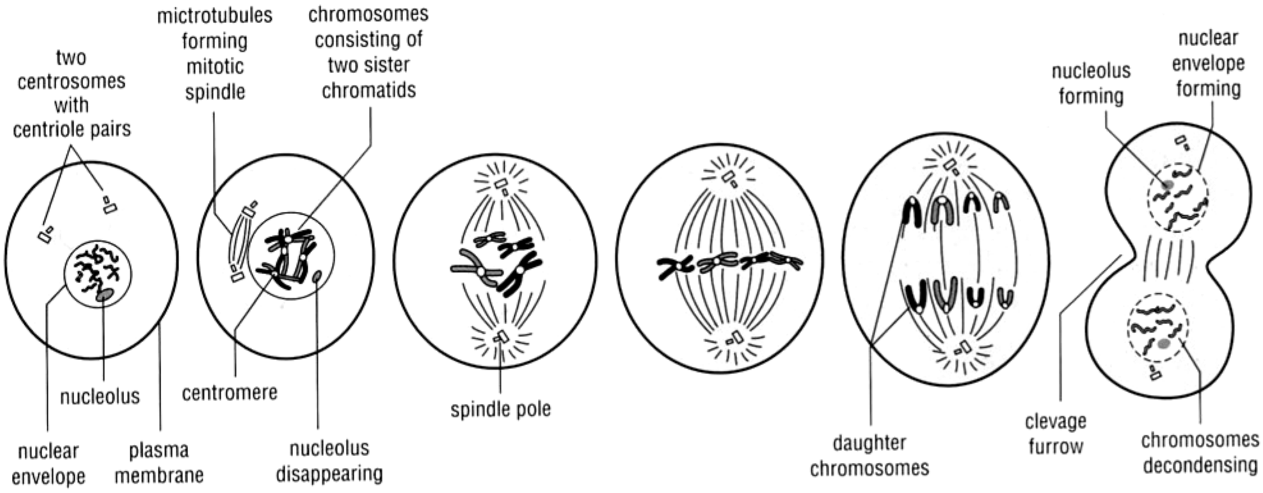
(c) What are the bases made of?

nitrogen - based

3. Define the following terms.

allele	Alternative forms of a gene. e.g. T, t
pure-bred	2 alleles for a gene are the same. e.g. TT, tt
hybrid	2 different alleles. e.g. Tt
homozygous	Pure-bred - alleles are the same. e.g. TT, tt
heterozygous	Hybrid - alleles are different. e.g. Tt
genotype	The genes for a characteristic. e.g. TT, Tt, tt
phenotype	How the organism looks. e.g. Tall, dwarf
sex-linked	Characteristic carried on the X or Y chromosome.
dominant	A gene that "masks" another. e.g. T dominates t
recessive	A gene that doesn't show if the dominant gene is present.
co-dominant	Both characteristics show. e.g. Red + white hair in cattle.
incomplete dominance	New characteristic shows - a "blend". e.g. Pink colour when red + white flowers are crossed.
haploid	Half the normal number of chromosomes.
gene	Information for a specific characteristic.

4. Below is a diagram showing a cell division process.



(a) Is this meiosis or mitosis?

mitosis

(b) Describe what is happening at each stage to the chromosomes.

1. Chromosomes become visible.
2. Chromosomes thicken and duplicate.
3. Chromosomes move towards the equator and spindle fibres form.
4. Chromosomes line up across the equator.
5. Spindle fibres pull the chromosomes apart.
6. Cell starts to "pinch off" to form two daughter cells.

(c) If possible, name each stage.

1 - 3: prophase

4: metaphase

5: anaphase

6: telophase

(d) In which cells does this process occur?

Normal cells - not reproductive cells.

5. (a) Why do cells undergo mitosis?

Need to replace damaged cells.

General growth - baby into toddler into teenager into adult.

(b) Why do cells undergo meiosis?

To halve the number of chromosomes in gametes ready for sexual reproduction.

(c) In what way do the cells produced by these two processes differ?

Mitosis - produces 2 daughter cells with diploid (normal) number of chromosomes.

Meiosis - produces 4 daughter cells with haploid (half) number of chromosomes.

6. Polled cattle do not have horns. This is a ***dominant*** phenotype arising from the action of an autosomal gene **P**. Horned cattle are therefore **pp** homozygotes.

(a) If a polled bull and a polled cow have a horned calf, what are the genotypes of the parents?

♂ Pp × ♀ Pp

Must have "p" present in each parent.

(b) If the same bull was bred with many horned cows, what would the proportions of polled and horned offspring overall be? ***Show full working***.

♂ Pp × ♀ pp

♀ \ ♂	P	p
p	Pp	pp
p	Pp	pp

GENOTYPE $\underbrace{Pp \quad Pp}_{50\%\ poll}$ $\underbrace{pp \quad pp}_{50\%\ horned}$

PHENOTYPE 50% poll 50% horned

7. In Andalusian chickens, the black (BB) characteristic is ***incompletely dominant*** to the white-splashed (WW) characteristic. The heterozygous (hybrid) condition produces blue chickens.

(a) How is incomplete dominance different to ***co-dominance***?

Incomplete - new or blended characteristic occurs. e.g. blue

Co-dominance - both colours show. e.g. black and white would show together.

(b) Determine the genotype and phenotype of the F1 generation from a pure-bred black Andalusian crossed with a blue Andalusian.

→ ♂ BB × ♀ BW

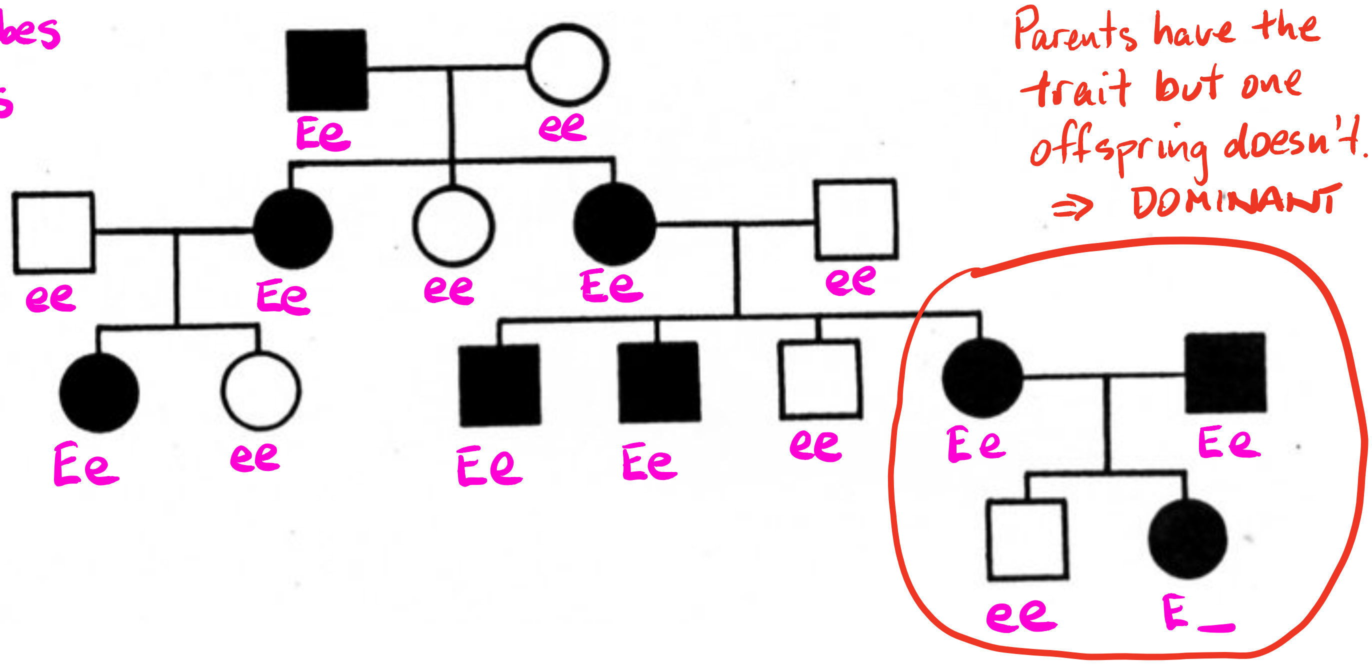
♀ \ ♂	B	B
B	BB	BB
W	BW	BW

GENOTYPE $\underbrace{BB \quad BB}_{50\%\ black}$ $\underbrace{BW \quad BW}_{50\%\ blue}$

PHENOTYPE 50% black 50% blue

8. The following pedigree shows the occurrence of no ear lobes in a family.

E = no lobes
e = lobes



Parents have the trait but one offspring doesn't.
⇒ DOMINANT

(a) Is the characteristic dominant or recessive (according to the pedigree)?

Dominant

(b) Explain why you made that determination.

See the explanation on the diagram.

(c) Determine the genotype of every individual in the pedigree. (Remember to start from the bottom!)

(d) How many females have no ear lobes?

5

9. (a) How many chromosomes are found in a human skin cell?

46

(b) How many chromosomes are found in the reproductive cells?

23

(c) Name the process used to make reproductive cells?

Meiosis

10. (a) What is the difference between the male and female sex chromosome pair?

XY (Y chromosome is very small) vs XX

(b) Who determines the sex of a child - the mother or father? Explain.

Father - he has the Y chromosome that determines maleness.

(c) Show, by punnet square, why half of any particular population would be expected to be female.

♂ XY × ♀ XX

♀ \ ♂	X	Y
X	XX	XY
X	XX	XY

GENOTYPE XX XX XY XY

PHENOTYPE 50% female 50% male

(d) A family has had five boys born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.

50 : 50 Half of the sperm in the father are Y chromosome.